PHILOSOPHY OF SCIENCE

Lecture 2: Kuhn’s Scientific Revolutions and Values in Science

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1924-1938: Vienna Circle

Popper

1950: Quine (Duhem)

1960:

Kuhn

1970: Science & technology studies, feminism

Explanation

1980: Postcolonialism

Constructive empiricism

THOMAS KUHN

- Not an ideal picture of science, but real science in its historical context

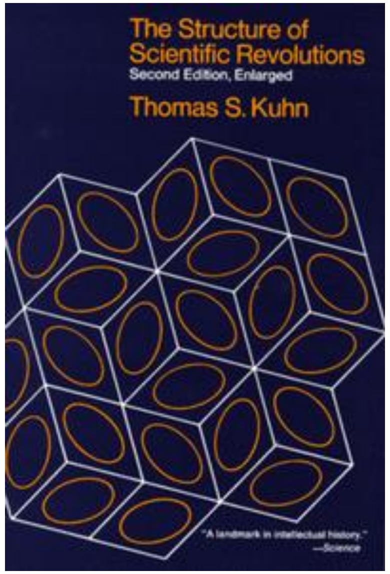
~~**Normative ideal of science**~~

~~**Justification of scientific knowledge**~~

How does science actually work?
What do scientists do?

- Kuhn one of the main opponents of logical empiricism
- Popper and Lakatos's work is in dialogue with Kuhn

THOMAS KUHN (1922-1996)



- Role of **history** in philosophy of science
- **Dynamical conception** of structure of science
- Critique of **scientific progress**
- Critique of the idea of a **theory-neutral observational language:** observation and description partly shaped by theoretical commitments

KUHN'S CRITIQUE

Positivist periodization

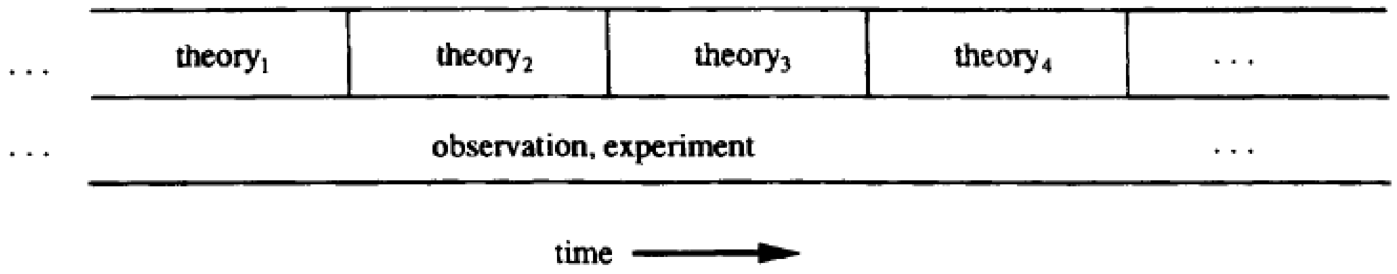


Figure 9.1 Positivist periodization. In the positivist periodization, the continuity and strength of the scientific picture issue from the accumulation of empirical results. Theory can and does change dramatically as needed to accommodate the new data.

Anti-positivist periodization (Kuhn)

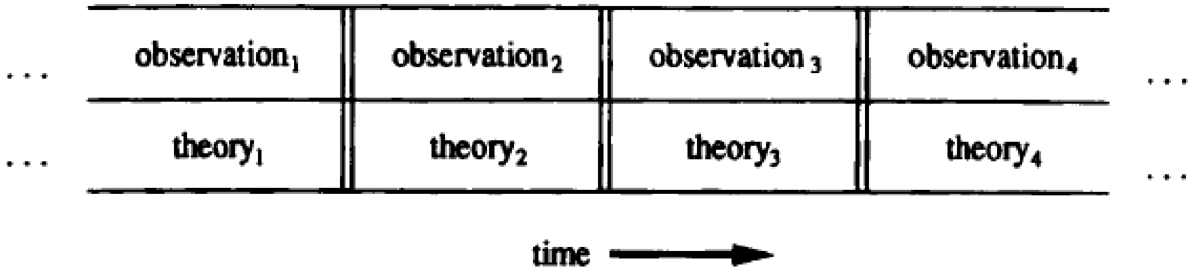


Figure 9.4 Antipositivist periodization. The antipositivist periodization inverts the positivist one in this respect: theory comes first. But now an additional assumption enters the picture, as theory and observation are assumed to be coperiodized—every powerful change in theory carries with it a concomitant shift in the standards of observation.

Galison (1997)

KUHN ON PHILOSOPHY OF SCIENCE

- SSR’s first sentence (Chapter 1: ‘A Role for History’):
    - **History**, if viewed as a repository for more than anecdote or chronology, could produce a decisive transformation in the image of science by which we are now possessed.
- Philosophers should study **real science** in its actual (historical) detail, not some idealized or simplified picture of it

KUHN'S MODEL OF SCIENTIFIC DEVELOPMENT

NORMAL SCIENCE	
paradigm	puzzles

↓

CRISIS	
discussion on paradigm	anomalies

↓

REVOLUTION	
new paradigm	puzzles

EXAMPLE: HISTORY OF CHEMISTRY

Preparadigmatic stage: alchemy	Joseph Priestley: phlogiston paradigm	Antoine Lavoisier: oxygen paradigm

WHAT IS A PARADIGM?

A framework for practising science, shared by the **scientific community** in a particular discipline

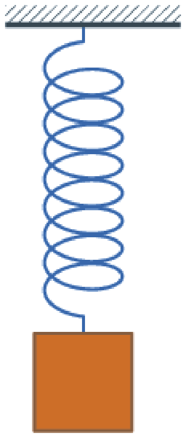
The paradigm holds the community together: science is a **social enterprise**

Elements: theoretical principles, methods of investigation, standards, tools, *exemplars*, metaphysical assumptions…

EXAMPLE: THE NEWTONIAN PARADIGM

$$\sum \mathbf{F} = m\mathbf{a}$$

- Theoretical principles: Newton’s laws, e.g. the second law
- Methods: *regulae philosophandi*
  - Tools: differential equations
- Exemplars: Earth-Sun system, Harmonic oscillator
- Metaphysical/philosophical assumptions:
  space and time (see Newton-Clarke/Leibniz debate)



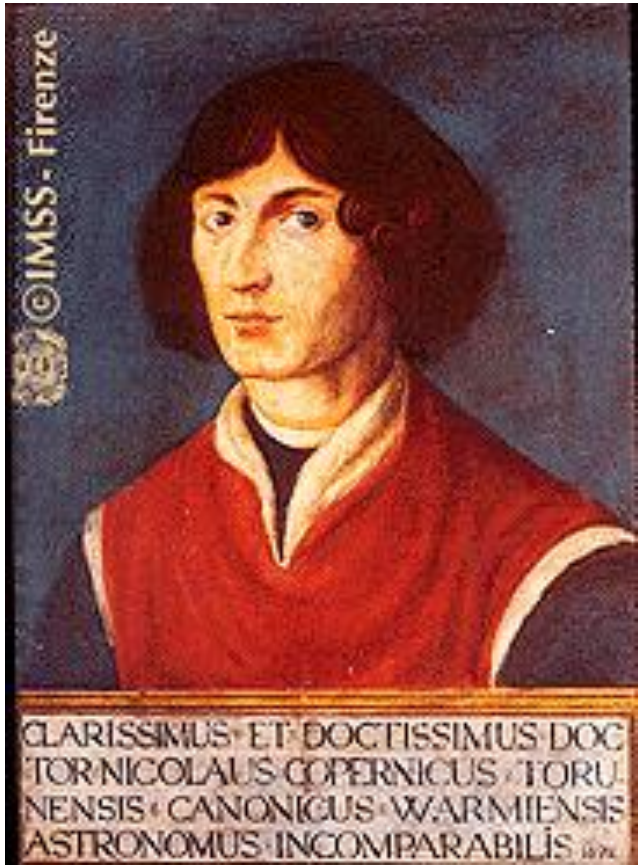
NORMAL SCIENCE

- **Solves puzzles within the paradigm**
- Paradigm is **accepted** unconditionally in normal science
  - Allows for progress
  - Critics of the paradigm marginalized or dispelled by the community
- Anomaly reflects on the scientist, not on paradigm
  - ‘Only a poor workman blames his tools’
- Normal science is **conservative** and seemingly **dogmatic** (cf. Popper)
- Astrology: similarities/differences Popper and Kuhn

SCIENTIFIC REVOLUTIONS

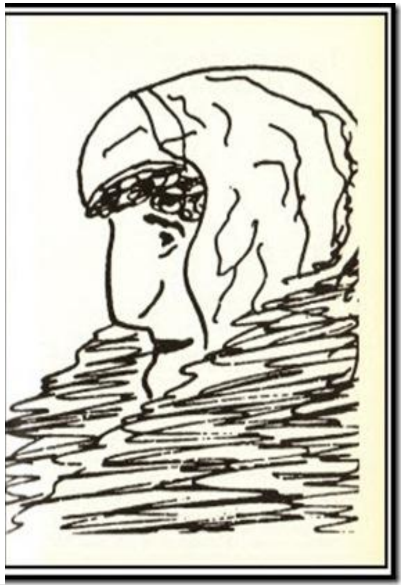
- **Anomaly:** puzzle resists solution, becomes anomaly
    - Inconsistent with understanding of paradigm
- **Crisis:** anomalies weaken confidence in paradigm (includes social and psychological dimensions)
- **Extraordinary science:** competing approaches explored, standards of assessment contested
- **Revolution**: victory of new paradigm

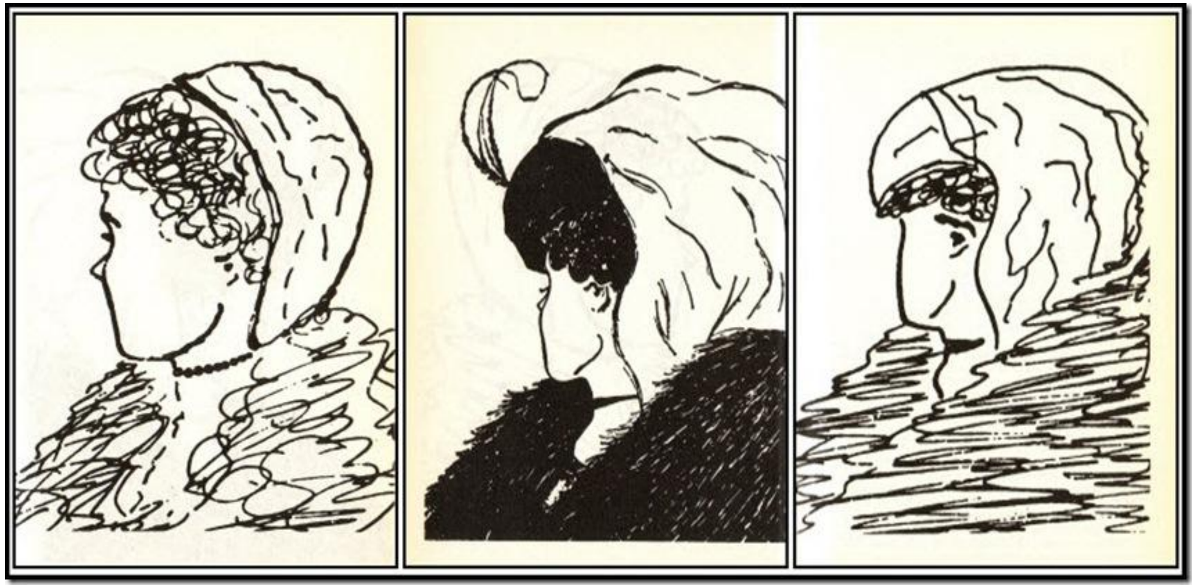
REVOLUTION IN ASTRONOMY



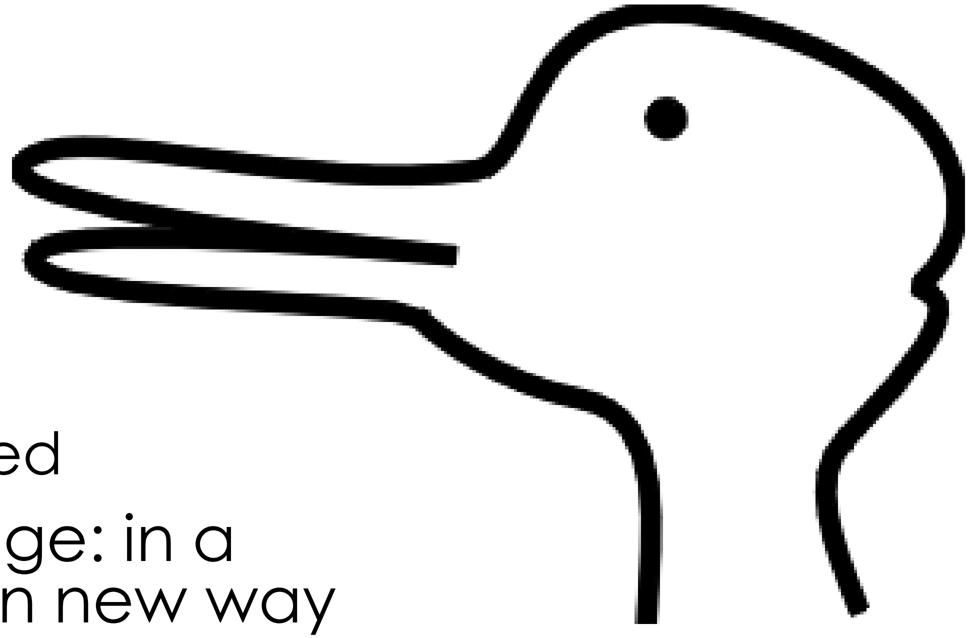
Ptolemy: Geocentric world-picture	Copernicus: Heliocentric world-picture

AN EXPERIMENT WITH PARADIGMS





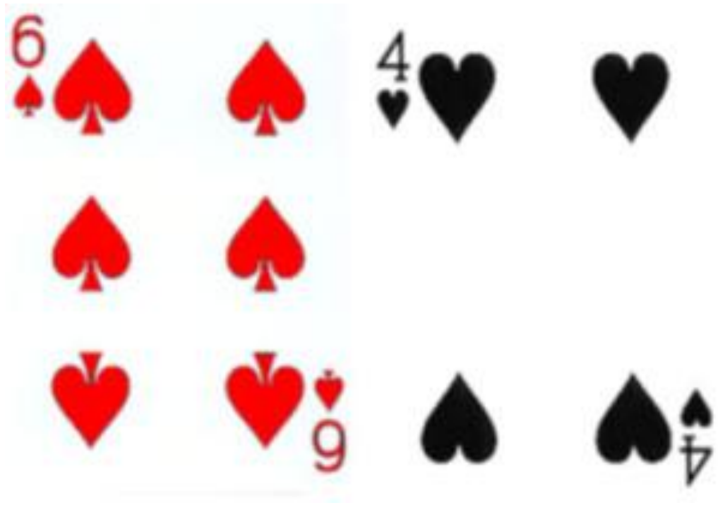
REVOLUTION: A GESTALT-SWITCH

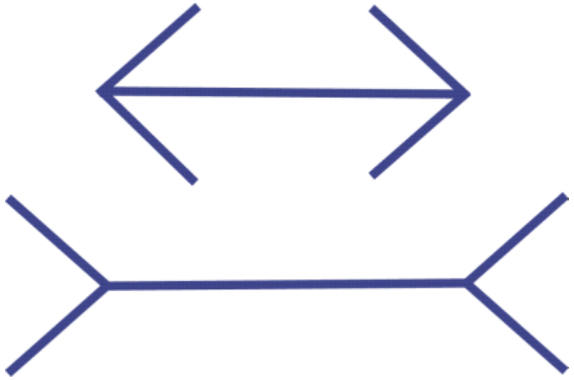


- Gestalt-psychology:
  - Humans observe patterns as wholes
  - Observation and interpretation are intertwined
- Switch between two ways of seeing the image: in a scientific revolution, scientists see the world in new way
- No fully theory-neutral description of observation: observation is **theory-laden** (paradigm-dependence)
- What scientists observe and what they take their observations to show is partly **shaped by their conceptual and practical commitments.**

OTHER EXAMPLES

- Playing cards experiment
- Müller-Lyer illusion

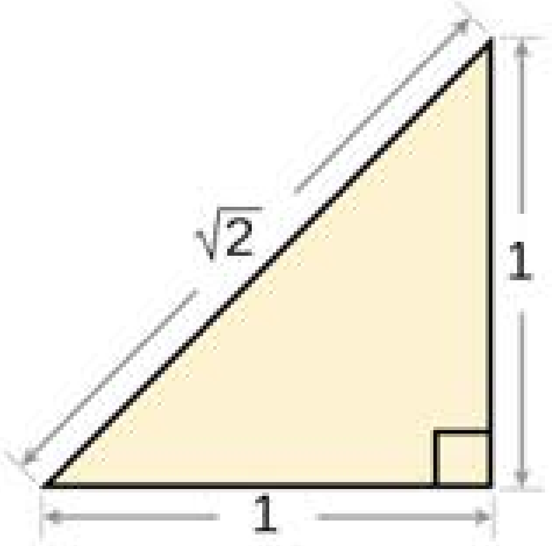




(See: Kahneman 2011, *Thinking, Fast and Slow*)

INCOMMENSURABILITY

- **Semantic incommensurability:** changes of meaning (language)
  - *'Mass' in Newtonian vs. Einsteinian paradigm*
  - *'Planet' before and after Copernicus*
- **Methodological incommensurability:** changes of standards
  - *Standards of explanation before and after Newton*
- **Problem:** how to account for progress in science?
- **Kuhn reformulated incommensurability as local untranslatability:** some terms cannot be translated without loss
  - There may be no translation that preserves all inferential relations



CRITIQUES OF KUHN

- "Winning" paradigm never eradicates all competitors
- No clear criterion of demarcation
- Metaphorical, unclear use of words
  - 21 different uses of the word 'paradigm' in a single book (Margaret Masterman, *The Nature of a Paradigm*, 1970)
- **Accusations of relativism:** no objective criteria to judge progress
- Accusations of analysing science as **irrational**
  - Accused by Lakatos of vindicating "mob psychology", because of comparison with political revolutions
- *The Copernican Revolution* (1957) stresses continuities of science much more than one might expect on the basis of *The Structure of Scientific Revolutions* (1962)

KUHN'S REPLY: VALUES AS CRITERIA OF CHOICE

- Kuhn denies relativism, insists that science is rational
- 'Objectivity, value judgment, and theory choice' (1974/7): **five shared values of theory appraisal** (objective criteria):

(1) Accuracy within its domain of application

(2) Consistency: internally and with other accepted theories

(3) Broad scope: consequences beyond observations for which designed

(4) Simplicity: brings order to phenomena

(5) Fruitfulness for new research findings

- Together, these criteria provide **the shared basis** for theory choice.
  - **Choice not algorithmic:** scientists who apply criteria may reach different conclusions
  - Previous experience as scientist counts. Mixture of shared and individual criteria
- Criteria are norms/values rather than rules.

ANNOUNCEMENTS

- Don’t forget to look at the **sample projects** on Canvas:
  - Files > Sample Projects
- Next week: **flipped classroom**
  - The lecture is available on Canvas
- Quine’s ‘On Empirically Equivalent Systems of the World’
  - Skip the discussion of William Craig’s work on pp. 324 (last paragraph) - 326 (1st paragraph)

VALUES IN SCIENCE

Externality (triptych) and internality models

Kuhnian values

Longino’s contextual empiricism and Heather Douglas

SCIENCE, FACTS, AND VALUES

Max Weber (1864-1920): Value-freedom of (social) science:

- Value relevance: values may guide selection of research problems
- Value freedom: scientific research should not present the researcher's own commitments as empirical conclusions
  - Once a question is selected, empirical claims should be assessed by evidential and methodological standards
- **Fact-value distinction:**
- Empirical statements and value judgements, or descriptive vs. normative statements

Compare with logical positivists:

SCIENCE, FACTS, AND VALUES

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Compare with logical positivists:

Similar distinction between factual and evaluative statements.

Values do not express propositions, but subjective attitudes

VALUE-FREE SCIENCE: TRIPTYCH-MODEL

- Also called: **externality model** (Heather Douglas)

Research agenda: problems, priorities, funding	Scientific research	Application of results and policy
May involve values	Value-free	May involve values
Panels 1 and 3: How to make the right choices? Panel 2: Can 'pure' science be value-free?		

VALUES *IN* SCIENCE? (PANEL 2)

- Logical positivists and Weber: no!
- Many argue that values can guide theory development / can help choose between theories:
- Epistemic values: accuracy, consistency, explanatory power, broad scope, simplicity, fruitfulness, maybe beauty
- Non-epistemic values: social desirability, consistency with political preferences, justice, etc.
- Question: Which of these are acceptable, and why?

KUHN: ‘OBJECTIVITY, VALUE JUDGMENT, AND THEORY CHOICE’

Kuhn (1977): ‘objective’ standards that provide ‘*the* shared basis for theory choice’:

- Accuracy, consistency, scope, simplicity, fruitfulness

However, these criteria:

- are individually *imprecise* → different applications
- may *conflict* with one another → different hierarchies

Therefore:

- scientists committed to the same criteria may end up making different choices (caused partly by contextual factors): no algorithmic procedure
- **rational disagreement,** required for progress, is possible and legitimate

CURRENT DEBATES ON VALUES IN SCIENCE

Since Kuhn's paper, philosophers of science have discussed values in various ways. Current key issues:

1. Distinction **epistemic/cognitive and non-epistemic** values
   Epistemic = conducive to cognitive aims of science, e.g. accuracy or simplicity
2. Inductive risk

THE WEAK VALUE-FREE IDEAL: EPISTEMIC VS. NON-EPISTEMIC VALUES

- Revised version of the triptych model
- Kuhn's (1977) values usually understood as epistemic

Research agenda: problems, priorities, funding	**Scientific research**	**Application of results and policy**
May involve non-epistemic values	Epistemic values	May involve non-epistemic values

Panels 1 and 3: How to make the right choices?

Panel 2: Can 'pure' science be value-free?

INTERNALITY POSITIONS: DOUGLAS AND LONGINO

Non-epistemic values can sometimes play a legitimate role *within* scientific reasoning

HELEN LONGINO'S CONTEXTUAL EMPIRICISM

- Data support a hypothesis only against background assumptions
  - Assumptions can include commmitments to non-epistemic values
- **Objectivity is achieved socially through transformative criticism**

Normative criteria for objectivity

= transformative dimension of critical discourse:

1. Recognized **avenues for criticism**
2. **Shared standards** that critics can invoke
3. Community is **responsive** to criticism
4. **Equality** of intellectual authority

Objectivity of science saved by social epistemology?

HEATHER DOUGLAS: INDUCTIVE RISK

- **Direct role of values:** impact of values on decisions/course of action
  - Direct role is compatible with externality picture
- **Indirect role:** when evidence is uncertain, the consequences of error may legitimately influence the **evidential threshold** for accepting/rejecting a claim. Values do not provide evidence for the hypothesis.
  - **Consequences of error:** require consideration of epistemic and non-epistemic values
  - **Internality picture:** non-epistemic values may indirectly affect scientific decisions under conditions of inductive risk

ROLE OF VALUES: COMPARISON

- Are values setting the **research agenda** or the applications?
  - ➢ Compatible with value-free ideal
- Are values setting a decision threshold under uncertainty?
  - Important in regulatory science, public health, environmental science, risk assessment
  - ➢ Compatible with Douglas' internality picture: indirect role
- Are values influencing the evidential connection between observations and hypotheses?
  - Because they enter the background assumptions that establish what counts as evidence or how data are interpreted:
  - Which variables are measured, which models are tested, which comparison groups are selected, what counts as explanation
  - ➢ Compatible with Longino's contextual empiricism

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CONCLUSION

- **Kuhn:** paradigms organise normal science.
  - Anomalies: crisis, extraordinary science, revolutions
- Challenges the idea that theory choice is governed by an algorithm or by theory-neutral observation
  - Scientific judgement depends on values, historical practices, and scientific communities
- This raises the question: **how can scientific judgement nevertheless be rational and objective?**
- Later debates distinguish:
  - **Epistemic values** in theory appraisal
  - **Non-epistemic values** in decisions under uncertainty
  - **Contextual values** in background assumptions that influence evidential reasoning

FURTHER READING

- Kuhn, *The Structure of Scientific Revolutions*
- Stanford Encyclopedia entry on Scientific Objectivity:
  - https://plato.stanford.edu/entries/scientific-objectivity/#EpisContValu